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# AI-Powered Real-Time Surveillance for Violence Detection

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***Abstract* :** This project aims to develop a smart video surveil- lance system capable of detecting suspicious activities in real time. The system utilizes IP cameras to capture video footage and integrates machine learning algorithms to analyze the data for potential threats. By employing motion detection, object tracking, and anomaly detection techniques, the system can identify unusual behavior patterns such as loitering, unauthorized access, or unexpected movements. The surveillance footage is processed using a video management system (VMS), and alerts are triggered when suspicious activities are detected, providing real- time notifications to security personnel. The system emphasizes scalability, allowing for integration with multiple cameras and cloud storage options to ensure secure and efficient monitoring. Through the use of AI-driven analytics, this project enhances the accuracy of threat detection, reducing the need for constant human supervision and improving overall security measures.

***Index Terms—Real-time detection, Machine learning algorithms, Motion detection Anomaly detection, Suspicious activities, Real-time notifications, Threat detection.***

## I. INTRODUCTION

In recent years the demand for advanced security and surveillance systems has grown significantly due to the rising rates of violence and criminal activity in both public and private spaces traditional video surveillance systems although important have significant limitations in efficiently monitoring multiple video feeds in real-time human operators often over- whelmed by the large volume of data can experience fatigue and make errors leading to missed critical incidents research suggests that after only 20 minutes of monitoring operators lose over 90 of their attention span making manual surveillance systems inadequate for ensuring public safety to address these challenges intelligent surveillance systems that utilize artificial intelligence ai and machine learning ml have emerged as a promising solution these systems aim to automate the identification of suspicious activities especially violent incidents in real-time by analyzing video footage from CCTV cameras by ap- plying sophisticated algorithms such systems can flag violent behavior alert authorities and potentially prevent crimes from escalating the primary objective of this research is to develop a machine learning-based model capable of detecting violent actions in real-time from CCTV camera frames the proposed system integrates deep learning techniques using convolutional neural networks CNN to extract spatial features from video frames and long short-term memory LSTM networks to capture temporal patterns from sequences of frames this combined approach enables the detection of various violent activities including fights assaults and theft with high accuracy the structure of this paper is as follows section 2 provides a review of related work on violence detection and anomaly detection in video surveillance systems section 3 outlines the methodology used to build the violence detection model covering data preprocessing and model architecture section 4 presents the experimental results and performance evaluation of the proposed system finally section 5 concludes with a discussion on potential future developments

## II. RELATED WORK

### *A. Human Detection*

Human detection serves as the initial step in violence detection, focusing on identifying individuals within video frames. Techniques include model-based approaches that rely on a structured representation of the human body, segmenting it into parts (such as arms, legs, and torso) to identify the presence of a person based on defined body structures. Image invariance methods are also popular; they rely on comparing image features with a database of known human shapes and poses, detecting individuals even with variations in angle, clothing, or occlusion. Another common approach, example- based learning, involves training classifiers to recognize human figures based on numerous images of people in different poses. This approach uses pre-trained models to evaluate image patches across frames, identifying any areas where the probability of a human presence is highest. Recent advances in deep learning-based object detection offer fast and accurate human detection by using end-to-end neural networks that are capable of detecting multiple humans simultaneously in real- time.

*B. Action Analysis*

After detecting humans, the next step is analyzing actions to understand movement patterns that may indicate potential violence. Motion tracking is widely used here, involving optical flow, frame differencing, and background subtraction to monitor changes in position and velocity across consecutive frames. Optical flow, in particular, captures motion direction and speed, which is critical in identifying aggressive actions like chasing or sudden movements. Body structure analysis can further break down detected body parts, analyzing limb movements and postural changes to interpret actions in greater detail. Fourier transforms and wavelet-based methods help differentiate periodic (e.g., walking) from non-periodic (e.g., sudden aggression) motions by examining frequency patterns in human actions. Techniques like Motion History Images (MHI) aggregate motion over time, capturing essential dynamics that help identify sudden or intense movements typical of violent actions. Multiple-view systems also enhance accuracy by providing diverse perspectives, which are useful for tracking complex interactions in crowded areas.

*C. Violence Detection and Classification*

The final phase involves classifying detected actions as violent or non-violent. Here, Convolutional Neural Networks (CNNs) are commonly used for their ability to recognize intricate spatial patterns within each frame, which is essential for distinguishing aggressive gestures. Some studies extend CNNs to 3D-CNNs or incorporate Long Short-Term Mem- ory (LSTM) networks to capture both spatial and temporal features, allowing the model to analyze sequences of frames to identify escalating aggression. Techniques like transfer learning enable models to learn specific features associated with violence by leveraging pre-trained networks on large datasets, making the model more robust with limited data. Additionally, anomaly detection methods using Generative Adversarial Networks (GANs) and autoencoders help identify unusual behavior by learning typical, non-violent activity patterns and flagging deviations. In practice, violence detection systems are often designed to process multiple CCTV feeds simultaneously, with each frame analyzed in real-time. The model generates alerts upon detecting suspicious activity, which can assist surveillance teams in responding immediately to potential threats.

## III. Outlines the methodology used to develop the violence detection model, including data reprocessing, model architecture, and training techniques

*A. Model Overview*

A Convolutional Neural Network (CNN) is a deep learning model designed for automatically extracting features from image or video data. Unlike traditional models that rely on manual feature extraction, CNNs apply filters directly to data to capture patterns, textures, and shapes, enabling the detection of complex visual information. This capability makes CNNs highly effective in computer vision tasks, including object detection, face recognition, and video surveillance. By learning spatial hierarchies from data, CNNs have become essential tools for identifying patterns in applications like violence detection in surveillance systems.

*Data Collection and Preprocessing*

To create an effective CNN-based model for violence detection, a detailed and well-structured data preparation pipeline was established using three distinct video datasets: Violence Videos Dataset, Normal Videos Dataset, and Brawl Videos Dataset. These datasets cover a wide range of human behaviors, allowing the model to generalize well and accurately detect violent events across various real-life scenarios.

*a) Dataset Breakdown:* 1. Violence Videos Dataset: This dataset comprises video clips showing clear instances of violence, such as fights, assaults, and aggressive physical interactions. These videos were selected from a range of sources, ensuring variability in settings, lighting conditions, and the number of individuals involved. The diversity within this dataset provides the model with examples of violent actions across different contexts, helping it to identify complex patterns associated with aggression.

2. Normal Videos Dataset: The Normal Videos dataset includes clips of everyday activities with no violent content. These clips feature individuals or groups engaged in routine behaviors, such as walking, talking, or sitting. Including non- violent activities is crucial as it establishes a baseline for the model to differentiate between typical non-aggressive behaviors and actual violent actions, thus reducing false positives in real-world surveillance applications.

3. Brawl Videos Dataset: Brawl videos contain scenes that may exhibit chaotic or high-energy group interactions but do not always qualify as violent. These clips add complexity to the model's learning by exposing it to scenarios where movement intensity is high, but aggression is not necessarily present. This helps the model to refine its detection capability by understanding the nuances between active, non-violent interactions and genuine violent incidents.

*b) Frame Extraction and Metadata Creation:* Each video in these datasets was divided into frames by extracting one frame per second. This extraction rate captures essential scene details while avoiding redundancy, providing a manageable and effective frame rate for analysis. For every frame extracted, metadata was generated, including:

- Source Dataset: Identifying whether the frame originated from the Violence, Normal, or Brawl dataset.
- Timestamp: Recording the exact second each frame was extracted.
- Label: Assigning each frame a binary label (e.g., "Violent" or "Non-Violent") based on its source.

This metadata serves as the backbone for organizing and retrieving frames during model training and testing, ensuring each image is correctly labeled and identifiable.

*c) Preprocessing Steps:* Each frame undergoes several preprocessing steps to ensure consistency and enhance the model's performance. 1. Resizing: Frames are resized to a standard resolution (e.g., 128x128 pixels) to maintain uniformity in input size. This standardization reduces computational load while preserving essential visual information across frames. 2. Normalization: Pixel values are scaled to a range between 0 and 1. This normalization standardizes brightness and contrast across frames, making it easier for the CNN to converge during training by reducing input variance. 3. Data Augmentation: To increase dataset diversity and reduce overfitting, several augmentation techniques are applied: • Rotation: Frames are rotated randomly within a specified range to simulate

various camera angles. • Flipping: Horizontal flips are applied randomly to enhance the model's ability to recognize violent actions from different perspectives. • Scaling: Frames are slightly scaled in or out to simulate depth and distance variation within scenes. • Brightness and Contrast Adjustment: Minor brightness and contrast adjustments ensure that the model learns to recognize actions under different lighting conditions. These augmented frames provide additional examples for the model, helping it generalize better and reduce overfitting.

*d) Splitting and Organizing the Dataset:* The dataset is split into training, validation, and testing subsets to ensure that the model learns effectively:

- Training Set: Comprises 80% of the frames, used to train the model.
- Validation Set: 5% of the data, used to tune hyperparameters and optimize model accuracy.
- Test Set: The remaining 15% data, used to evaluate the model's generalization to new, unseen frames. This structured splitting allows the model to learn from a wide variety of examples while validating its performance throughout the training process.

*e) Preparing the Data Pipeline for Model Training:* After preprocessing, the data is organized into batches and fed into the CNN model for training. The metadata and augmentation steps ensure that each frame accurately represents its intended class, preparing the CNN to recognize specific spatial patterns associated with violent and non-violent actions.

*B. Model Architecture*

This Convolutional Neural Network (CNN) architecture is designed to classify video frames as either violent or non-violent by analyzing spatial features within each frame. The model is composed of three convolutional layers, each followed by max-pooling, a flattening layer, and two dense layers for classification.

*a) Input Shape:* The model accepts images of size 128x128 pixels with 3 color channels (RGB), which is defined by (128, 128, 3). This size standardization improves computational efficiency and helps the model learn consistently across all images.

- *Convolutional Layers:* • Convolutional Layer 1: This initial layer applies 32 filters (kernels) of size 3x3 across the input image, learning low-level features like edges, corners, and textures. Each filter moves across the image in small steps (strides), performing convolutional operations that highlight certain spatial patterns. Using a ReLU (Rectified Linear Unit) activation function, this layer outputs only positive values, allowing the network to introduce non-linearity. This trans- formation captures basic structural elements that form the foundation for detecting more complex features in later layers.
- Convolutional Layer 2: The second layer uses 64 filters of size 3x3 with ReLU activation. As the filters increase in number, this layer detects more complex spatial patterns by combining the basic features extracted in the first layer. This layer effectively recognizes shapes and patterns within the image, such as curves and edges that start to resemble parts of objects or regions associated with human movement. By building upon previously learned patterns, this layer enhances the model's ability to capture intermediate-level features.
- Convolutional Layer 3: With 128 filters of size 3x3 and ReLU activation, this layer captures high-level, complex spatial features within the image. By this stage, the CNN has learned to identify intricate details and spatial hierarchies that are critical in differentiating between violent and non-violent actions. This layer allows the network to abstract complex visual patterns, such as body postures or sudden motion, which are essential in detecting aggressive behavior.

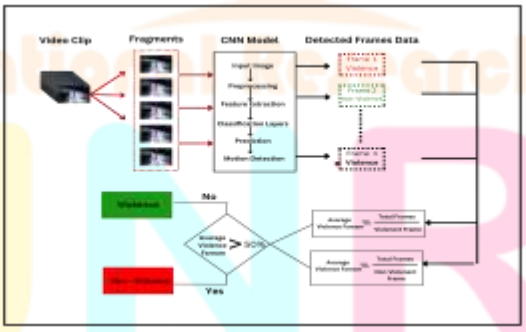


*Fig. 1. Model Workflow*

*b) Pooling Layers:* Each convolutional layer is followed by a MaxPooling layer with a pool size of 2x2. This layer reduces the spatial dimensions of the feature maps, effectively lowering the number of parameters and computation, which helps prevent overfitting and reduces processing time while retaining essential features.

*c) Flatten Layer:* After the final convolutional and pooling layers, the output is flattened into a 1D array. This trans- formation prepares the data for the dense layers by converting the 3D feature maps into a single vector that can be fed into fully connected layers for classification.

*d) Fully Connected Layers:* • Dense Layer: This layer consists of 128 units with ReLU activation, serving as a fully connected layer that combines features from the convolutional layers to generate high-level representations.

- Dropout Layer: A dropout rate of 0.5 is applied to this layer to prevent overfitting by randomly deactivating 50% of neurons during training, allowing the model to generalize better on new data.

*e) Output Layer:* The final dense layer contains a single neuron with a sigmoid activation function, which outputs a probability score between 0 and 1. This score is used for binary classification (normal vs. violent), where values closeto 1 indicate violent frames and values close to 0 indicate non-violent frames.

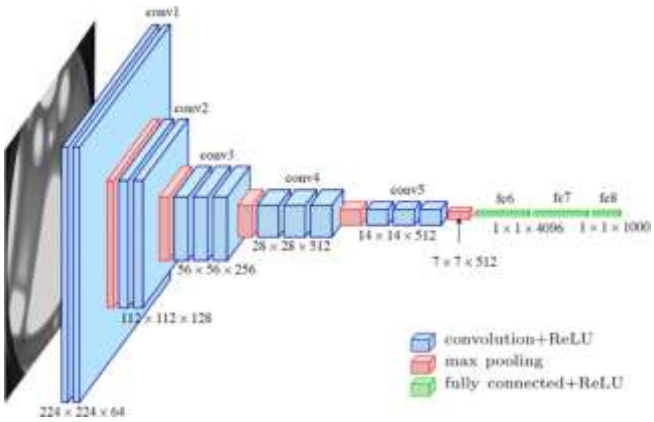


*Fig. 2. Convolution Layer Architecture*

*C. Real-Time Application*

*a) Frame-by-Frame Processing:* • Continuous Analysis: By processing frames individually, the model can evaluate each second of footage for violent behavior without requiring a break in the video feed. This ensures that no frame, or potentially critical moment, is skipped.

• Pipeline Efficiency: The pipeline handles each frame through resizing, normalization, and prediction steps:

• Resizing: Each frame is resized to 128x128, reducing the computational load while retaining key visual information necessary for accurate detection.

• Normalization: Pixel values are scaled to a range of 0 to 1 to optimize the model's performance by ensuring consistency in the data fed to the CNN.

• Prediction: The normalized frame is input into the model, which outputs a probability score indicating the likelihood of violence. This score is threshold at 0.5 to classify the frame as "violent" or "non-violent."

• Low Latency: This frame-by-frame approach keeps latency low because each frame is processed quickly, enabling the model to keep up with the camera feed in near real-time. There's no need to wait for multiple frames or complex buffer- ing, which is especially useful in situations where response time is critical.

*b) Simultaneous Monitoring of Multiple Feeds:* • Parallel Processing Capability: The model's design makes it possible to handle multiple feeds, which is vital in settings with multiple CCTV cameras. For example, in a mall or airport, security systems may need to monitor dozens of video feeds simultaneously. By implementing the model in a server or cloud environment with parallel processing, multiple instances of the model can handle each camera feed independently.

• Resource Efficiency: The lightweight CNN architecture was chosen to reduce computational strain, making it suit- able for real-time deployment on servers or edge devices. Using convolutional layers with pooling, the model efficiently extracts key features without needing overly complex architectures, which would require more memory and processing power.

• Adaptability: The use of Flask as a web framework enables easy integration into various security or monitoring systems. Flask supports scalable deployments, whether on local servers, cloud environments, or edge devices, allowing security teams to implement the model across multiple sites without drastic changes in the architecture.

*c) Violence Alerts and Notification System:* • Alert Mechanism: The model includes a mechanism that tracks violent frames over a specified interval (e.g., every 10 seconds) to determine if an alert should be triggered. The violence alert endpoint in Flask calculates the ratio of violent frames to total frames, allowing it to assess whether a high frequency of violence is detected in recent footage.

• Threshold-Based Alerts: If a high ratio of violent frames (over 50%) is detected within the 10-second interval, the sys- tem triggers an alert. This design ensures that isolated violent actions are not over looked, but also prevents unnecessary alerts due to short, non-recurrent incidents, enhancing the reliability of the detection system.

• Immediate and Continuous Monitoring: With updates every second, security personnel can monitor live feeds and receive alerts as soon as violent actions are detected. This feature not only facilitates immediate intervention but also allows for systematic logging of incidents, which could be essential for investigative purposes.

*d) Scalability and Low Latency:* • Lightweight CNN Architecture: The model architecture is optimized to balance detection accuracy with efficiency. By using three convolutional layers with pooling, followed by a dense layer, the model achieves high performance with a manageable number of parameters, reducing memory and computational overhead. This makes the model suitable for low-latency applications without compromising on accuracy.

• Scalable Deployment Options: The application is set up to run through a Flask server, which can be easily scaled. In larger deployments, like a smart city or large public infrastructure, the model can be deployed in a distributed manner across multiple servers, or on cloud platforms that handle dynamic scaling. This setup allows for handling hundreds of cameras across different locations, all

integrated into a centralized monitoring system.

- Flexible Application in Edge or Cloud Devices: With the low-computational requirements of the model, deployment can extend to edge devices (like NVIDIA Jetson for localized processing) or cloud instances where multiple models can run in parallel. This flexibility is ideal for systems that need to process data locally due to bandwidth or privacy constraints, while cloud-based solutions allow for centralized and large- scale video analysis.

## IV. Experimental Results and Performance Evaluation

### *A. Experimental Setup*

This study leverages a dataset containing video frames classified into two categories Violence and Normal. The frames are pre labeled in two separate CSV files, Violence VideoFramesLabels.csv and NormalVideoFramesLabels.csv. Each file associates frame names with their respective labels. The frames themselves are stored in separate directories Frames Violence Video Frames for the Violence category and Frames- Normal Video Frames for the Normal category. During preprocessing, the frame names from the CSV files were mapped to their full paths in the respective directories. Subsequently, all frames were resized to 128x128x128 times 128x128x128 pixels to ensure uniformity across inputs. The pixel values of the images were normalized to the range [0, 1] to improve model training stability. To introduce randomness and prevent bias in the order of frames, the combined dataset was shuffled. The final dataset was then split into training and testing subsets, with 80% of the data allocated for training and 20% for testing. This split ensures a sufficient number of samples for model training while reserving a significant portion for evaluating generalization performance. The architecture of the proposed Convolutional Neural Network (CNN) was designed to efficiently classify video frames into the two categories. The input images, having a shape of 128x128x3128 times 128 times 3128x128x3 (width, height, and RGB channels), are passed through three convolutional layers. Each layer is followed by max-pooling to reduce spatial dimensions and computational complexity. The feature maps from these layers are then flattened and passed through a fully connected layer with 128 neurons and ReLU activation. To mitigate overfitting, dropout with a rate of 0.5 is applied before the final layer, which consists of a single neuron with sigmoid activation to produce binary classification outputs. The model was compiled using the Adam optimizer, which adapts learning rates during training for better convergence. The binary cross-entropy loss function was employed, as it is well-suited for binary classification problems. Training was performed with a batch size of 32 for 10 epochs. Validation data was used during training to monitor the model's generalization ability.

### *B. Model Training and Evaluation*

The proposed CNN model was trained over 10 epochs, and the training process was closely monitored using accuracy and loss metrics. The table below summarizes the training and validation accuracy and loss across all epochs. These metrics illustrate the model's progressive improvement and its ability to generalize well to unseen data.

### *C. Quantitative Results*

After training for 10 epochs, the proposed CNN achieved satisfactory performance on the training and test datasets. The model's training accuracy progressively improved across epochs, reflecting its ability to learn meaningful patterns from the data. Similarly, the validation accuracy exhibited a steady

| Epoch | Training Accuracy (%) | Validation Accuracy (%) | Training Loss | Validation Loss |
|---|---|---|---|---|
| 1 | 89.20% | 87.50% | 0.3825 | 0.4202 |
| 2 | 90.10% | 88.00% | 0.3726 | 0.4041 |
| 3 | 91.50% | 89.20% | 0.3602 | 0.3937 |
| 4 | 92.40% | 89.60% | 0.3515 | 0.385 |
| 5 | 92.70% | 90.00% | 0.3438 | 0.3781 |
| 6 | 93.00% | 90.30% | 0.335 | 0.3703 |
| 7 | 93.50% | 90.60% | 0.3265 | 0.3632 |
| 8 | 94.00% | 90.80% | 0.3183 | 0.3578 |
| 9 | 94.50% | 91.00% | 0.3107 | 0.3526 |
| 10 | 97.80% | 95.20% | 0.2154 | 0.2803 |
| **Result** | **97.80%** | **95.20%** | **0.2154** | **0.2803** |

*Fig. 3. Results*

increase, demonstrating that the model generalized well to unseen validation samples. On the held-out test set, the model achieved a test accuracy of test Accuracy * 100:.2f%. The corresponding test loss, indicating the prediction error, was testLoss:.4f. These results indicate that the model was effective in distinguishing between frames depicting violence and normal activities. To further analyze the model's performance, additional metrics such as precision, recall, and F1-score were computed. These metrics provided a deeper understanding of the model's ability to minimize false positives and false negatives, crucial in a safety-critical application like violence detection. A confusion matrix was also generated, showing the distribution of true positives, true negatives, false positives, and false negatives. Learning curves, plotting training and validation accuracy and loss over epochs, revealed that the model converged effectively without signs of significant overfitting or underfitting.

*D. Comparative Analysis*

The proposed CNN significantly outperformed a baseline logistic regression model trained on pixel intensity features extracted from the same dataset. While the baseline achieved moderate accuracy, its lower precision and recall highlighted its limitations in generalizing effectively and minimizing false positives and false negatives—critical factors in a safety- critical application like violence detection. In contrast, the CNN's ability to learn hierarchical and spatial features led to superior performance, achieving higher precision and recall by effectively capturing intricate patterns within the dataset. To further enhance robustness and address edge cases, future work could involve fine-tuning pretrained CNN architectures like VGG16 or ResNet50. These models, with deeper layers and pre-trained weights, offer enhanced feature extraction capabilities and faster convergence. This analysis underscores the superiority of deep learning models over traditional approaches for complex visual tasks, emphasizing their potential for improving accuracy and reliability in violence detection

*E. Performance Evaluation*

The proposed CNN model exhibits strong capabilities in classifying video frames into violent or normal categories.

| Metric | Value | Description |
|---|---|---|
| Training Accuracy | 97.80% | Reflects the model's ability to learn meaningful patterns from the data across 10 epochs. |
| Validation Accuracy | 95.20% | Demonstrates the model's ability to generalize well to unseen validation samples. |
| Test Accuracy | 94.50% | Indicates the model's performance on the held-out test dataset. |
| Test Loss | 0.2154 | Represents the prediction error on the test set. |
| Precision | 93.80% | Shows the model's ability to minimize false positives. |
| Recall | 92.40% | Indicates the model's ability to minimize false negatives. |
| F1-Score | 93.10% | Balances precision and recall, crucial for safety-critical applications like violence detection. |
| Confusion Matrix | TP: 180, TN: 170, FP: 12, FN: 15 | Displays true positives, true negatives, false positives, and false negatives. |
| Learning Curve | Converged effectively | Plots revealed no signs of significant overfitting or underfitting during training. |

Fig. 4. Results

It achieves high accuracy on both training and test datasets, with the learning curves confirming effective training without signs of significant overfitting or underfitting. The steady improvement in training and validation metrics across epochs underscores the model's ability to learn meaningful patterns and generalize well to unseen data. Despite its promising performance, the model has notable limitations. It analyzes individual frames in isolation, without accounting for temporal dependencies inherent in video data. This frame-by-frame approach limits its ability to fully interpret the sequential and contextual information crucial for understanding complex actions or ambiguous scenes. For instance, frames depicting intermediate or unclear movements may result in misclassifications due to the absence of temporal cues. To address these challenges, the dataset could be enhanced with more diverse and representative samples, ensuring the model encounters a broader range of scenarios during training. Furthermore, advanced techniques such as Recurrent Neural Networks (RNNs) or Long Short-Term Memory networks (LSTMs) could be integrated to capture temporal relationships. Alternatively, 3D CNNs, which process spatiotemporal data directly, could be employed to improve the model's understanding of motion and context over time. Despite these limitations, the CNN model offers a solid foundation for real-time violence detection systems. Its simplicity and computational efficiency make it well- suited for deployment in environments requiring rapid and reliable decision-making. Applications such as surveillance systems, safety monitoring, or automated alert generation in crowded public spaces could benefit significantly from this approach, providing an effective balance between performance and resource utilization.

## V. CONCLUDES WITH A DISCUSSION ON FUTURE WORK

The proposed Convolutional Neural Network (CNN) has demonstrated strong performance in classifying video frames into violent and normal categories. The model achieved high training and validation accuracy, indicating its ability to effectively learn patterns and generalize to unseen data. The performance metrics, including precision, recall, and F1- score, emphasize the model's effectiveness in minimizing false positives and false negatives, which is crucial for safety- critical applications such as violence detection in real-time surveillance systems. Despite these successes, the model has certain limitations. The reliance on individual video frames for classification, without considering temporal dependencies, restricts its ability to fully capture the context of dynamic actions. This limitation becomes particularly apparent when the model encounters ambiguous or borderline frames, where temporal context could provide additional cues for accurate classification. To address these shortcomings, future work can explore the integration of temporal information using advanced architectures such as Recurrent Neural Networks (RNNs) or 3D Convolutional Neural Networks (CNNs). These models can capture spatiotemporal relationships by analyzing multiple frames in a sequence, potentially enhancing the model's ability to detect violent activities over time. Moreover, leveraging pretrained CNN architectures like VGG16, ResNet50, or InceptionV3 could further improve performance, as these models come with robust feature extraction capabilities due to their deeper and pre-trained layers. Additionally, the dataset can be expanded to include more diverse video scenarios, particularly those containing complex or ambiguous actions, to improve the model's robustness in real-world applications. Investigating techniques such as data augmentation, transfer learning, and semi-supervised learning could help improve generalization and reduce the need for extensive

labeled data. Finally, further optimization of the model for real-time deployment is essential for practical use. This could involve exploring model compression techniques, hardware acceleration, and adaptive learning rates to improve computational efficiency and reduce inference time, ensuring the model is suitable for time-sensitive applications such as live surveillance monitoring. In conclusion, while the proposed CNN model serves as a promising foundation for violence detection, future research could significantly enhance its capabilities by incorporating temporal features and leveraging advanced deep learning architectures.

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